1.1) $\int \left(\sqrt{x}+\frac{1}{x}\right)dx = \ln|x| + \frac{2}{3}x\sqrt{x} + c$ ✓

1.2) $\int \left(x^2 - x + \frac{2}{\sqrt[4]{x}}\right)dx = \frac{x^3}{3} - \frac{x^2}{2} + \frac{8}{3}\sqrt[4]{x^3} + c$ ✓

1.3) ~~$\int \frac{\sqrt{x}+2}{x}dx$~~ $\to \frac{x}{2} + 2\sqrt{x} + c$ ✗ → $2\sqrt{x} + 2\ln|x| + c$

1.4) $\int (x - \sin x)\,dx = \frac{1}{2}x^2 + \cos x + c$ ✓

1.5) $\int \left(x^4 + \sqrt{x} - \sqrt[5]{x-1}\right)dx = \frac{1}{5}x^5 + \frac{2}{3}x^{3/2} - \frac{5}{6}\sqrt[5]{(x-1)^6} + c$ ✓

1.6) $\int \left(x^{1/4} - x^{1/3} + \cos x\right)dx = \frac{4}{5}x^{5/4} - \frac{3}{4}x^{4/3} + \sin x + c$ ✓

1.7) $\int \frac{3}{1+x^2}dx = 3\arctan x + c$ ✓

1.8) $\int (x-1)^3\,dx = \frac{1}{4}(x-1)^4 + c$ ✓ 12/15

1.9) $\int \frac{\sqrt{x}+\sqrt[3]{x}}{\sqrt[4]{x}}dx \to \int \frac{\sqrt{x}}{\sqrt[4]{x}}dx + \int \frac{\sqrt[3]{x}}{\sqrt[4]{x}}dx$ → $\frac{4}{5}\sqrt[4]{x^5} + \frac{12}{13}\sqrt[12]{x^{13}} + c$ ✓

1.10) $\int \frac{\sqrt[3]{x}+2x+6}{\sqrt{x}}dx = \int \frac{\sqrt[3]{x}}{\sqrt{x}} + \frac{2x}{\sqrt{x}} + \frac{6}{\sqrt{x}}\,dx$ → $\frac{6\sqrt[6]{x^7}}{7} + \frac{4\sqrt{x^3}}{3} + 12\sqrt{x} + c$ ✓

1.11) ~~$\int (e^x + 3^x + 5^{-x})\,dx \to$?~~ → $e^x + \frac{3^x}{\ln(3)} + \frac{(5^{-x})}{\ln(5)} + c$ ✓

1.12) $\int (e^{x-1} + e^x + e^{x+1})\,dx \to e^{x-1} + e^x + e^{x+1} + c$ ✓

1.13) $\int \frac{x^3+3x^2+1}{\sqrt[4]{x^3}}dx \to \int x^{\frac{9}{4}}dx + 3\int x^{\frac{5}{4}}dx + \int \frac{1}{\sqrt[4]{x^3}}dx$ → $\frac{4}{21}\sqrt[4]{x^{21}} + \frac{12}{17}\sqrt[4]{x^{17}} + 4\sqrt[4]{x} + c$ ✓

1.14) $\int (x^2 + \sin x - \cos x)\,dx = \frac{1}{3}x^3 - \cos x - \sin x + c$ ✓

1.15) ~~$\int \left(\sqrt{x-1} \pm \sqrt{x+1}\right)dx$~~ → $\frac{2}{3}\left(\sqrt{(x-1)^3} + \sqrt{(x+1)^3}\right) + c$ ✓

2.1) $f(x) = \begin{cases} \sin x & \text{se } x \le 0 \\ x^2 & x > 0 \end{cases}$ → $F(x) = \begin{cases} -\cos x + c_1 & x \le 0 \\ \frac{x^3}{3} + c_2 & x > 0 \end{cases}$ → $\begin{cases} -\cos x + 1 & x \le 0 \\ \frac{x^3}{3} & x > 0 \end{cases}$

$\lim_{x\to 0^-} F_1(x) = -1 + c_1$, $\lim_{x\to 0^+} F_2(x) = c_2$ → $c_1 = c_2 + 1$; $c_2 = c_1 \to 0$

2.2) $f(x) = \begin{cases} \cos x & \text{se } x \le 0 \\ e^x & \text{se } x > 0 \end{cases}$ → $F(x) \begin{cases} \sin x + c_1 & x \le 0 \\ e^x + c_2 & x > 0 \end{cases}$ → $\begin{cases} \sin x & x \le 0 \\ e^x & x > 0 \end{cases}$

$c_1 = 1 + c_2 \to 1 + c_2 = 1 + c_2$

2.3) $f(x) = \begin{cases} 0 & x \le 0 \\ \sqrt{x} & x > 0 \end{cases}$ → $F(x) = \begin{cases} x + c_1 & x \le 0 \\ \frac{2\sqrt{x^3}}{3} + c_2 & x > 0 \end{cases}$ → $\begin{cases} x & x \le 0 \\ \frac{2\sqrt{x^3}}{3} & x > 0 \end{cases}$

$c_1 = c_2$

2.4) $f(x) = \begin{cases} |x+1| & x < 0 \\ |x^2 - 5x + 1| & x \ge 0 \end{cases}$ → $\begin{cases} x+1 & -1 \le x < 0 \\ -x-1 & x < -1 \end{cases}$

$x_{1,2} = \frac{5 \pm \sqrt{21}}{2}$ → $\begin{cases} x^2 - 5x + 1 & 0 \le x \le \frac{5-\sqrt{21}}{2} \\ -x^2 + 5x - 1 & \frac{5-\sqrt{21}}{2} < x \le \frac{5+\sqrt{21}}{2} \\ x^2 - 5x + 1 & x > \frac{5+\sqrt{21}}{2} \end{cases}$

→ $\begin{cases} -x - 1 & x < -1 \\ x + 1 & -1 \le x < 0 \\ x^2 - 5x + 1 & 0 \le x \le \frac{5-\sqrt{21}}{2} \\ -x^2 + 5x - 1 & \frac{5-\sqrt{21}}{2} < x \le \frac{5+\sqrt{21}}{2} \\ x^2 - 5x + 1 & x > \frac{5+\sqrt{21}}{2} \end{cases}$ → $F(x) = \begin{cases} -\frac{x^2}{2} - x + c_1 & x < -1 \\ \frac{x^2}{2} + x + c_2 & -1 \le x < 0 \\ \frac{x^3}{3} - \frac{5x^2}{2} + x + c_3 & 0 \le x \le \frac{5-\sqrt{21}}{2} \\ -\frac{x^3}{3} + \frac{5x^2}{2} - x + c_4 & \frac{5-\sqrt{21}}{2} < x \le \frac{5+\sqrt{21}}{2} \\ \frac{x^3}{3} - \frac{5x^2}{2} + x + c_5 & x > \frac{5+\sqrt{21}}{2} \end{cases}$

$\lim_{x\to -1^-} F(x) = \frac{1}{2} + c_1$, $\lim_{x\to -1^+} F(x) = -\frac{1}{2} + c_2$

$\lim_{x\to 0^-} F(x) = c_2$, $\lim_{x\to 0^+} F(x) = c_3$

$\lim_{x\to \frac{5-\sqrt{21}}{2}^-} F(x) = \frac{(5-\sqrt{21})^3}{24} - \frac{(5-\sqrt{21})^3}{8} + \frac{5-\sqrt{21}}{2} + c_4$

$\lim_{x\to \frac{5-\sqrt{21}}{2}^+} F(x) = \frac{(5-\sqrt{21})^3}{24} - \frac{3(5-\sqrt{21})^2}{24} + \frac{12(5-\sqrt{21})}{24} + c_5$

$\frac{(5-\sqrt{21})\left[(5-\sqrt{21})^2 - 3(5-\sqrt{21}) + 12\right]}{24} + c_5 = \frac{(5-\sqrt{21})\left[25+21-10\sqrt{21}-15+3\sqrt{21}+12\right]}{24}$

$(5-\sqrt{21})(43-7\sqrt{21})$

$(5-\sqrt{21})(43-7\sqrt{21})$

$(5-\sqrt{21})(43-7\sqrt{21})$

$(362 - 78\sqrt{21})$

2.5) $f(x) = \begin{cases} x+2 & x < -1 \\ x^2 & -1 \le x \le 1 \\ \sqrt[4]{x} & x > 1 \end{cases}$ → $F(x) = \begin{cases} \frac{x^2}{2} + 2x + c_1 & x < -1 \\ \frac{x^3}{3} + c_2 & -1 \le x \le 1 \\ \frac{4}{5}\sqrt[4]{x^5} + c_3 & x > 1 \end{cases}$ → $F(x) = \begin{cases} \frac{x^2}{2} + 2x + C & x < -1 \\ \frac{x^3}{3} - \frac{7}{6} + C & -1 \le x \le 1 \\ \frac{4\sqrt[4]{x^5}}{5} - \frac{49}{30} + C & x > 1 \end{cases}$

$\lim_{x\to -1^-} F(x) = -\frac{3}{2} + c_1$, $\lim_{x\to -1^+} F(x) = -\frac{1}{3} + c_2$ → $c_2 = c_1 - \frac{7}{6}$; $c_1 = c_2 + \frac{9-2}{6} = c_2 + \frac{7}{6}$

$\lim_{x\to 1^-} F(x) = \frac{1}{3} + c_2$, $\lim_{x\to 1^+} F(x) = \frac{4}{5} + c_3$ → $c_2 = \frac{12-5}{15} + c_3 = \frac{7}{15} + c_3$; $c_3 = c_2 - \frac{7}{15} = c_1 - \frac{7}{6} - \frac{7}{15}$

$c_1 - \frac{35+14}{30} = c_1 - \frac{49}{30}$

2.6) $f(x) = \begin{cases} -x^{-1} & x \le -1 \\ \sqrt{x+2} & x \ge -1 \end{cases}$ → $F(x) = \begin{cases} -\frac{1}{\ln|x|} + c_1 & x \le -1 \\ \frac{2\sqrt{(x+2)^3}}{3} + c_2 & x \ge -1 \end{cases}$ → $\begin{cases} -\frac{1}{\ln|x|} + \frac{2}{3} + C & x \le -1 \\ \frac{2\sqrt{(x+2)^3}}{3} + C & x \ge -1 \end{cases}$

$\lim_{x\to -1^-} -\frac{1}{\ln|x|} = 0 + c_1$, $\lim_{x\to -1^+} \frac{2\sqrt{(x+2)^3}}{3} = \frac{2}{3} + c_2$ → $c_1 = \frac{2}{3} + c_2$

2.7) $f(x) = \sqrt{|x|}$ → $\begin{cases} \sqrt{-x} & x < 0 \\ \sqrt{x} & x \ge 0 \end{cases}$ → $F(x) = \begin{cases} -\frac{2\sqrt{x^3}}{3} + C & x < 0 \\ \frac{2\sqrt{x^3}}{3} + C & x \ge 0 \end{cases}$

$\lim_{x\to 0^-} -\frac{2\sqrt{x^3}}{3} + c_1 = c_1$, $\lim_{x\to 0^+} \frac{2\sqrt{x^3}}{3} + c_2 = c_2$

2.8) $f(x) = e^{|x+2|}$ → $\begin{cases} e^{-(x+2)} & x < -2 \\ e^{x+2} & x \ge -2 \end{cases}$ → $F(x) = \begin{cases} -e^{-(x+2)} + 2 + C & x < -2 \\ e^{x+2} + C & x \ge -2 \end{cases}$

$F(x) = \begin{cases} -e^{-(x+2)} + c_1 & x < -2 \\ e^{x+2} + c_2 & x \ge -2 \end{cases}$ → $\lim_{x\to -2^-} -1 + c_1$, $\lim_{x\to -2^+} 1 + c_2$ → $-1 + c_1 = 1 + c_2$

2.9) $f(x) = \begin{cases} \sin x & x < 0 \\ e^{x+1} & x \ge 0 \end{cases}$ → $F(x) = \begin{cases} -\cos x + c_1 & x < 0 \\ e^{x+1} + c_2 & x \ge 0 \end{cases}$ → $\lim_{x\to 0^-}$, $\lim_{x\to 0^+}$